

Distinct GNS classes have no unitary intertwiner

`MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation`

theorem

The class index makes the selected pure-GNS family pairwise unitarily inequivalent.

Statement

Let A be a unital complex C^* -algebra, with the positive order used for its states, and fix a pure state ω . A pure state is an extreme point of the normalized positive continuous linear functionals. Let I be the set of pure-state GNS representations modulo unitary equivalence. For each $j \in I$, choose a pure state φ_j in that class, retaining $\varphi_o = \omega$ literally at $o = [\omega]$. Write (H_j, π_j, ξ_j) for its GNS space, representation and canonical cyclic vector. If $i \neq j$, no surjective complex-linear isometry $E : H_i \rightarrow H_j$ intertwines the two selected representations. Explicitly, for every such E , the equations $E\pi_i(a) = \pi_j(a)E$ cannot hold for all $a \in A$.

Assumptions

The GNS classes $i, j \in I$ are assumed to be distinct. A candidate E is a unitary map between the two complete Hilbert spaces; it need not carry ξ_i to ξ_j . The same representative selection based at ω is used on both sides.

Conclusion

The selected representations form a family with no unitary intertwiners between distinct indices. This supplies precisely the pairwise-inequivalence hypothesis of the generic atomic-shell construction.

Two distinct pure states can belong to the same GNS class. The assertion here is about distinct classes. Vanishing of arbitrary bounded intertwiners requires an additional irreducibility argument and is not the conclusion of this declaration.

Proof route

An intertwining unitary would certify GNS equivalence of the two selected states. Taking their quotient classes would therefore identify i and j , contradicting their assumed distinctness.

▼ Proof steps

1. Assume that the candidate unitary E satisfies the intertwining equations. The selected representations are exactly the GNS actions of φ_i and φ_j , so E witnesses their GNS equivalence.
2. The cited representative-class theorem says that GNS-equivalent selected representatives have equal indices: each chosen representative has the class it was chosen from. Applying that theorem gives $i = j$, contrary to the hypothesis.

Main citations

- The stated existence or structural result —
MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation
- Selected concrete GNS representations —
MathlibAnnex.CStarAlgebra.PureState.selectedRepresentation
- Equivalent selected representatives have the same class —
MathlibAnnex.CStarAlgebra.PureState.representative_injective_on_classes
- Each representative lies in its specified class —
MathlibAnnex.CStarAlgebra.PureState.classOf_representative

Supporting route explanation

Lean source signature (exact)

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theorem no_unitaryIntertwiner_selectedRepresentation (root : PureState A)
  {i j : GNSClass A} (hij : i ≠ j) (e : SelectedGNS root i ≈i[ $\mathbb{C}$ ] SelectedGNS root j) :
  ¬ StarAlgHom.Intertwines (selectedRepresentation root i)
    (selectedRepresentation root j)
    (e : SelectedGNS root i →L[ $\mathbb{C}$ ] SelectedGNS root j)
```

In the source	Mathematical meaning
root : PureState A; i j : GNSClass A	Fix the root pure state ω of A and two indices in the set I of pure-GNS unitary-equivalence classes. Each has selected data (H_i, π_i, ξ_i) or (H_j, π_j, ξ_j) .
hij : i ≠ j	The GNS classes are distinct, a stronger condition than merely unequal state functionals.
e : SelectedGNS root i ≈ _i [$\mathbb{C}$] SelectedGNS root j	A candidate surjective complex-linear isometry $E : H_i \rightarrow H_j$, already a unitary between these Hilbert spaces.
(e : SelectedGNS root i → _L [$\mathbb{C}$] SelectedGNS root j)	The same unitary E read as a bounded complex-linear map, without changing its direction or values.
¬ StarAlgHom.Intertwines (selectedRepresentation root i) (selectedRepresentation root j) ...	The equations $E\pi_i(a)x = \pi_j(a)Ex$ cannot hold for every $a \in A$ and $x \in H_i$. No condition $E\xi_i = \xi_j$ is imposed. The theorem excludes unitary intertwiners, not all bounded maps.

Further source notes: In the linked proof, representative_injective_on_classes converts that unitary equivalence into equality of the class indices; it is separately cited.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: MathlibAnnex.CStarAlgebra.PureState.no_unitaryIntertwiner_selectedRepresentation

Accepted content SHA-256: d05909d87e1bf86548852e6e5126f3e546c42bbbb21dac861bb8bbfebd012a23

Accepted source guide SHA-256: ddb408f27abc434f4a84e440e167ab826e8bd98d4a32bbb591497bb1836cd8b0

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73